

HIGH-FREQUENCY STATISTICAL ARBITRAGE: COINTEGRATION ANALYSIS OF BTC/ETH



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1. EXECUTIVE SUMMARY

This project evaluates the viability of a Pairs Trading (Statistical Arbitrage) strategy applied to Bitcoin (BTC) and Ethereum (ETH) using high-frequency (1-minute) market data. The objective was to exploit temporary pricing inefficiencies between these two highly correlated assets.

While the assets demonstrated a correlation coefficient of **0.98**, the **Augmented Dickey-Fuller (ADF)** test failed to reject the null hypothesis of non-stationarity $p = 0.47$. Consequently, the mean-reversion strategy underperformed, resulting in a net loss. This result highlights the critical distinction between correlation and cointegration in quantitative finance.

2. METHODOLOGY

2.1 HYPOTHESIS: LAW OF ONE PRICE

Pairs trading relies on the assumption that two assets with similar fundamental drivers will share a long-term equilibrium. Deviations from this equilibrium (the "Spread") are assumed to be temporary and mean-reverting.

2.2 MATHEMATICAL MODEL

1. **Hedge Ratio (β):** Calculated using Ordinary Least Squares (OLS) regression on the training set (first 60% of data).

$$P_{ETH} = \beta \cdot P_{BTC} +$$

Result: $\beta = 0.0423$ (implying a hedge of ~0.042 BTC for every 1 ETH).

2. **The Spread (S_t):**

$$S_t = P_{ETH,t} - (0.0423 \times P_{BTC,t})$$

3. **Signal Generation (Z-Score):** To normalize volatility, we calculate the rolling Z-Score of the spread:

$$Z_t = \frac{S_t - \mu_{60}}{\sigma_{60}}$$

- **Short Signal:** $Z_t > 2.0$ (Spread is too high; Sell ETH, Buy BTC).

- **Long Signal:** $Z_t < -2.0$ (Spread is too low; Buy ETH, Sell BTC).

3. DATA ANALYSIS

3.1 CORRELATION SYNCHRONIZATION

We utilized 1-minute interval data for the period of Jan 10, 2026 – Jan 17, 2026. As shown in Figure 1, the assets move in near-perfect lockstep.

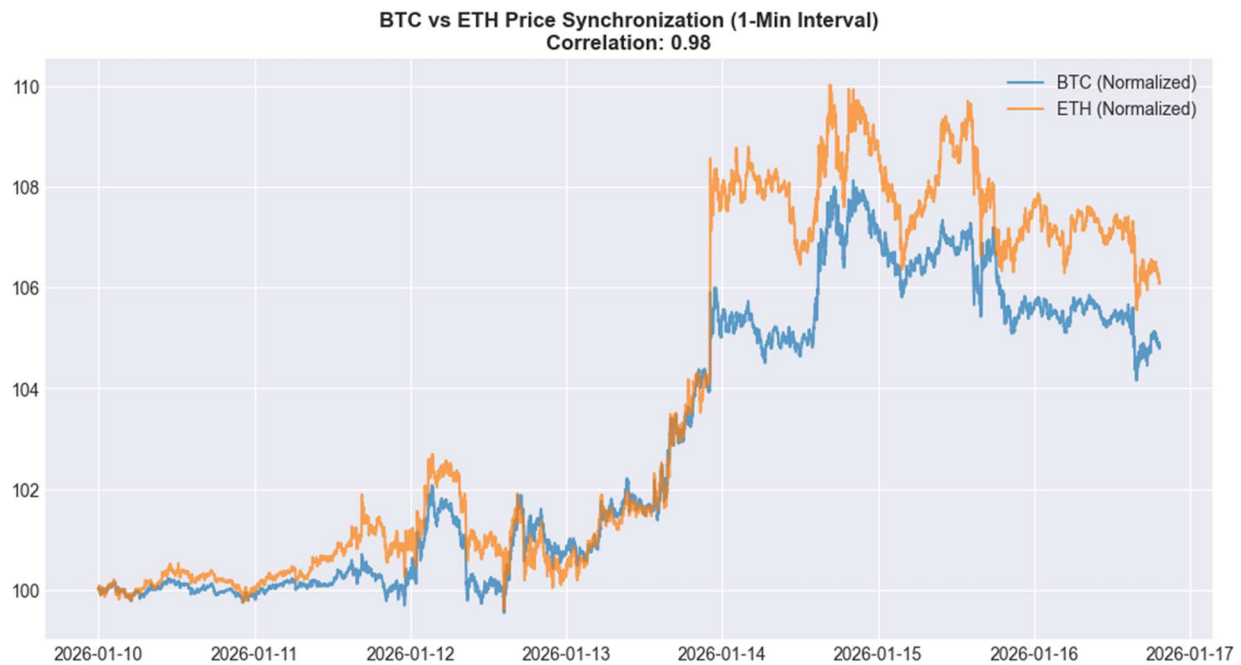


Figure 1: Normalized price action of BTC and ETH showing 0.98 correlation.

3.2 COINTEGRATION TESTING (THE WARNING SIGN)

Before executing the strategy, we performed the **Augmented Dickey-Fuller (ADF)** test to verify stationarity.

- **Null Hypothesis (H_0):** The spread has a unit root (is non-stationary).
- **Result:** P-Value = **0.477**.
- **Conclusion:** We failed to reject H_0 . The spread was following a random walk rather than mean-reverting. This was a critical risk indicator suggesting the strategy would likely fail in this specific regime.

4. BACKTEST RESULTS

4.1 SIGNAL FREQUENCY

Figure 2 displays the Z-Score dynamics. The strategy generated frequent entry signals as the spread drifted continuously beyond the 2-sigma threshold, indicating a structural "break" in the relationship rather than a temporary divergence.

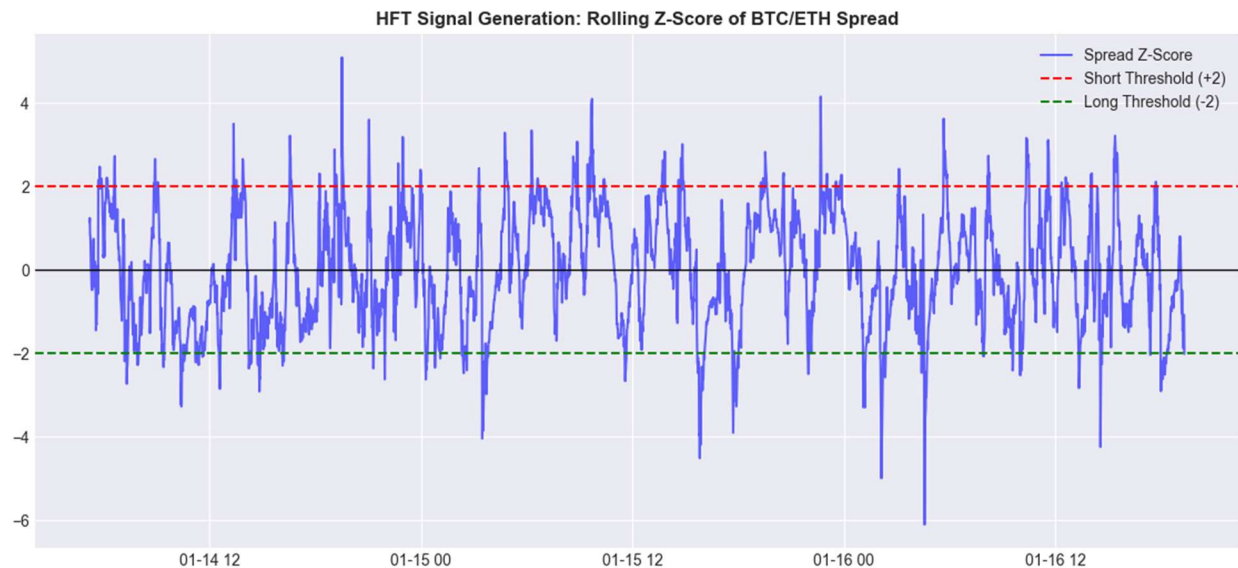


Figure 2: Rolling Z-Score of the spread. Note the persistent deviation away from zero.

4.2 PERFORMANCE ATTRIBUTION

The strategy suffered a consistent drawdown, ending with a loss of **-\$398.88**. Because the spread was non-stationary, it did not revert to the mean after opening positions; instead, it continued to widen, accumulating losses on open trades.

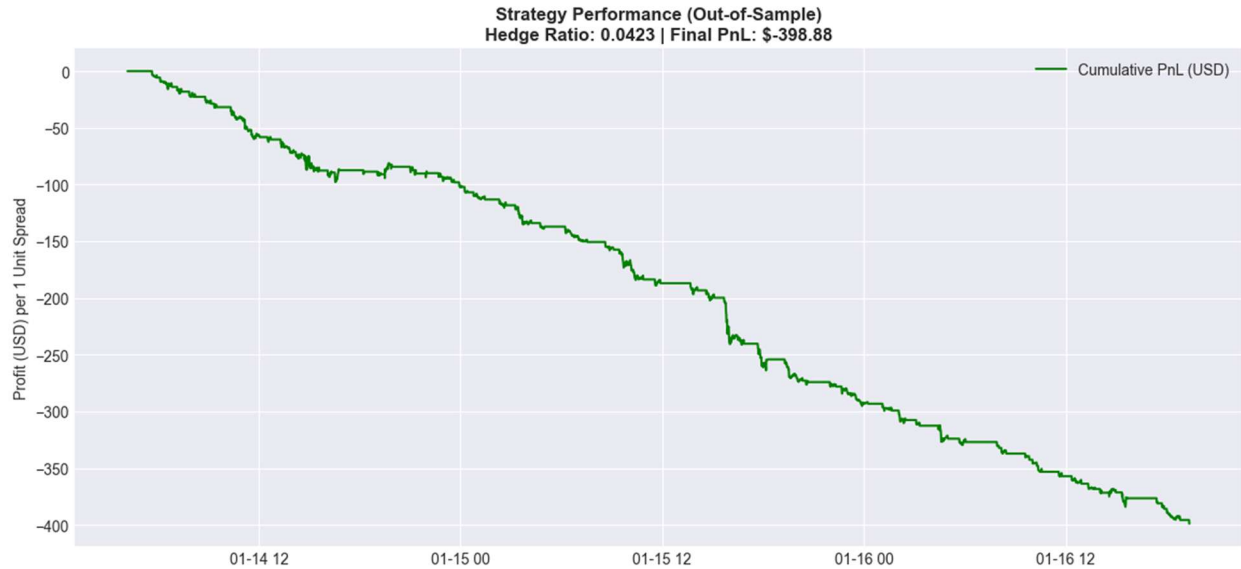


Figure 3: Cumulative Profit/Loss. The downward slope confirms the "fighting the trend" behavior of a mean-reversion strategy in a momentum regime.

5. CONCLUSION & RISK MANAGEMENT

This project serves as a rigorous negative proof. It demonstrates that **correlation is a necessary but insufficient condition for pairs trading**.

- **Quantitative Lesson:** A robust algo-trading engine must include a "Regime Filter." If $P_{ADF} > 0.05$, the system should automatically halt trading, as the fundamental assumption of mean reversion is violated.
- **Market Insight:** During the tested week (Jan 2026), ETH and BTC decoupled structurally, likely due to idiosyncratic news events affecting one asset more than the other, breaking the cointegration.